

Grounding DINO

Initial Findings

James Corino · August 2026



Why Grounding DINO?

1

Open vocabulary

Detects objects from a natural-language prompt at inference time — no fixed class list, no retraining.

2

Deep vision–language fusion

Text guides the detector throughout the pipeline, not just at a final classification step.

3

Tunable output

Box threshold filters candidate boxes; text threshold controls prompt-word matching in labels.

ROLE IN OUR PIPELINE

Stage 1 — Grounding DINO
locate faces / facial regions



Stage 2 — dedicated forensic model
classify manipulation on cropped faces

Setup: official Swin-T checkpoint · macOS /
Apple M2, CPU-only · transformers pinned to
4.37 · ~3.4 s per image

Experiment Design

21 test images



Frontal — 10

LFW (HF mirror)

4 categories



Multiple — 4

stock photos

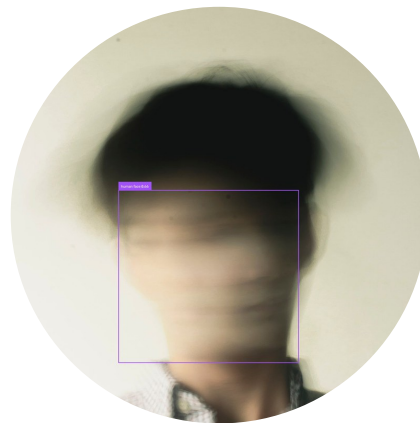
5 prompts



Profile / angled — 5

stock photos

3 threshold settings



Occluded / blurred — 2

stock photos

Prompts: "human face" · "person" · "face . eyes . mouth" · "real face" · "manipulated face" ; Batch script for saving annotated images + JSON (boxes, phrases, scores, per-image time).

Prompt Behavior

“face . eyes . mouth” multiplies boxes by design — 3–11 boxes/image vs. 1–4 for “human face”: three grounding targets, not higher sensitivity.

“real face” / “manipulated face” add no forensic signal — box counts track “human face” almost exactly (ex: [18,3,5,6] vs [17,3,5,6]); the model matches on “face”, so these prompts stay exploratory.

“person” ≠ “human face” — identical on profile/occluded (1 box each), but diverges with scene complexity: 10 vs 18 boxes on the densest crowd, 2.3 vs 1.6 on single faces.

Detection held up on profile & occluded faces — stable 1 box/image across face prompts.



“human face”
1 box

“face . eyes . mouth”
3 targets

*Same image, same thresholds ;
only the prompt changes.*

Threshold Sensitivity

No single threshold works everywhere ; raising it removes real faces, not just noise.



box = 0.25

3 boxes

box = 0.35

1 box

box = 0.45

0 boxes — face missed

Profile / occluded image across the sweep

Crowded scene: 26 → 18 → 1 boxes — 0.45
collapses a genuinely crowded scene to one face.

Frontal image: extra boxes were real background faces — not duplicates; a global cutoff discards genuine detections.

Implication: per-scene threshold policy or NMS-style post-processing needed before feeding a forensic model. Default 0.35/0.25 is a reasonable middle ground.

Limitations & Next Steps

KNOWN LIMITS

- No deepfake content tested yet — FF++ access pending
- Small categories: occluded n=2, multiple n=4
- Small / low-resolution faces not yet represented
- No universal threshold; duplicates need post-processing

NEXT

- Re-run “real/manipulated face” prompts on deepfake frames
- Expand occluded & multiple-face sets; add small/low-res condition
- Evaluate NMS / per-scene thresholds as pre-filter for the forensic classifier

Takeaway: Grounding DINO reliably localizes faces across pose and occlusion, but thresholding alone can't separate noise from real faces, and its prompts carry no indicator.